1) List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

Main Components Found Inside a Desktop PC Cabinet

Motherboard
The main circuit board of the computer that connects all hardware components and allows them to communicate with each other.

CPU (Central Processing Unit)
The processor that performs calculations and executes instructions for running programs and operating the system.

CPU Cooler/Heatsink
Attached to the CPU to dissipate heat and prevent the processor from overheating.

RAM (Random Access Memory)
Temporary memory used to store data and programs that are currently being accessed by the CPU.

Graphics Card (GPU)
Responsible for rendering images, videos, and graphics. It can be integrated into the CPU or installed as a dedicated card.

SMPS (Switched Mode Power Supply)
Converts AC power from the wall outlet into DC power required by the motherboard, CPU, storage drives, and other components.

Hard Disk Drive (HDD)
A storage device that uses magnetic disks to store the operating system, software, and user files.

Solid State Drive (SSD)
A fast storage device that uses flash memory instead of moving parts, providing quicker boot and loading times.

Cooling Fans
Maintain airflow inside the cabinet and help keep components at safe operating temperatures.

CMOS Battery
A small battery on the motherboard that preserves BIOS settings and system date/time when the computer is powered off.

SATA Cables
Data cables used to connect HDDs, SSDs, and optical drives to the motherboard.

Cabinet/Case
The outer enclosure that houses and protects all internal computer components.

2) Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

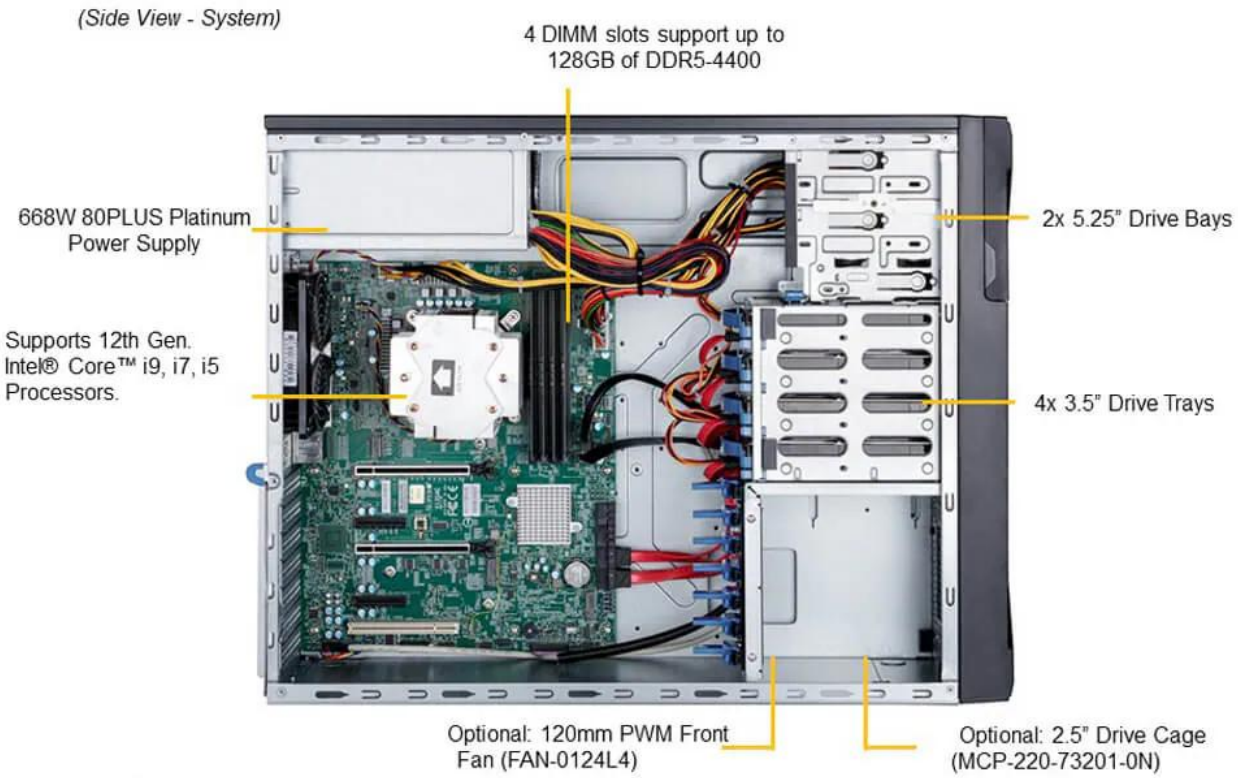
Desktop PC Disassembly Report

Step 1: Shut Down the Computer and Unplug Power
The desktop computer was completely shut down. The power cable, monitor cable, keyboard, mouse, and all external devices were disconnected.

Step 2: Remove the Side Panel

The side panel screws were removed using a screwdriver. The side cover was then slid backward and lifted away from the cabinet.

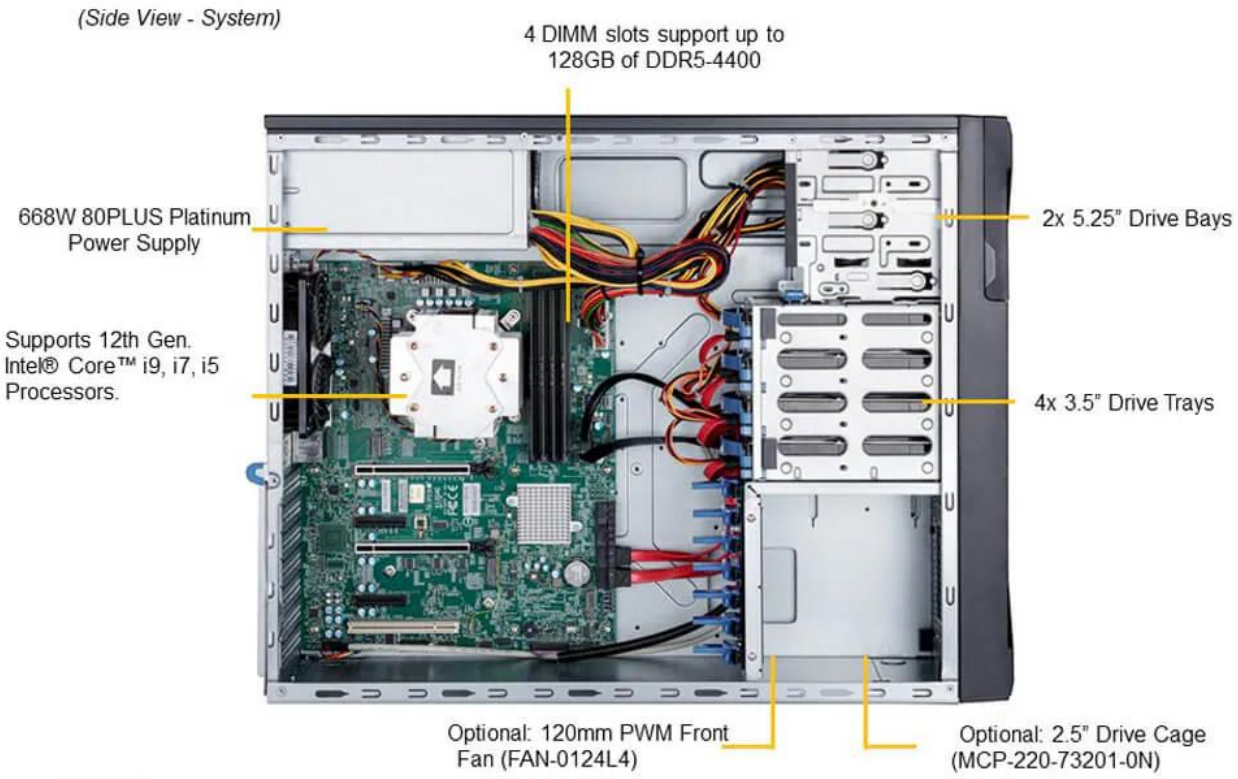
Diagram/Photo Placeholder:



Step 3: Identify Internal Components

The major internal components including the motherboard, CPU cooler, RAM, storage devices, SMPS, and cables were identified.

Diagram/Photo Placeholder:



Components Identified:

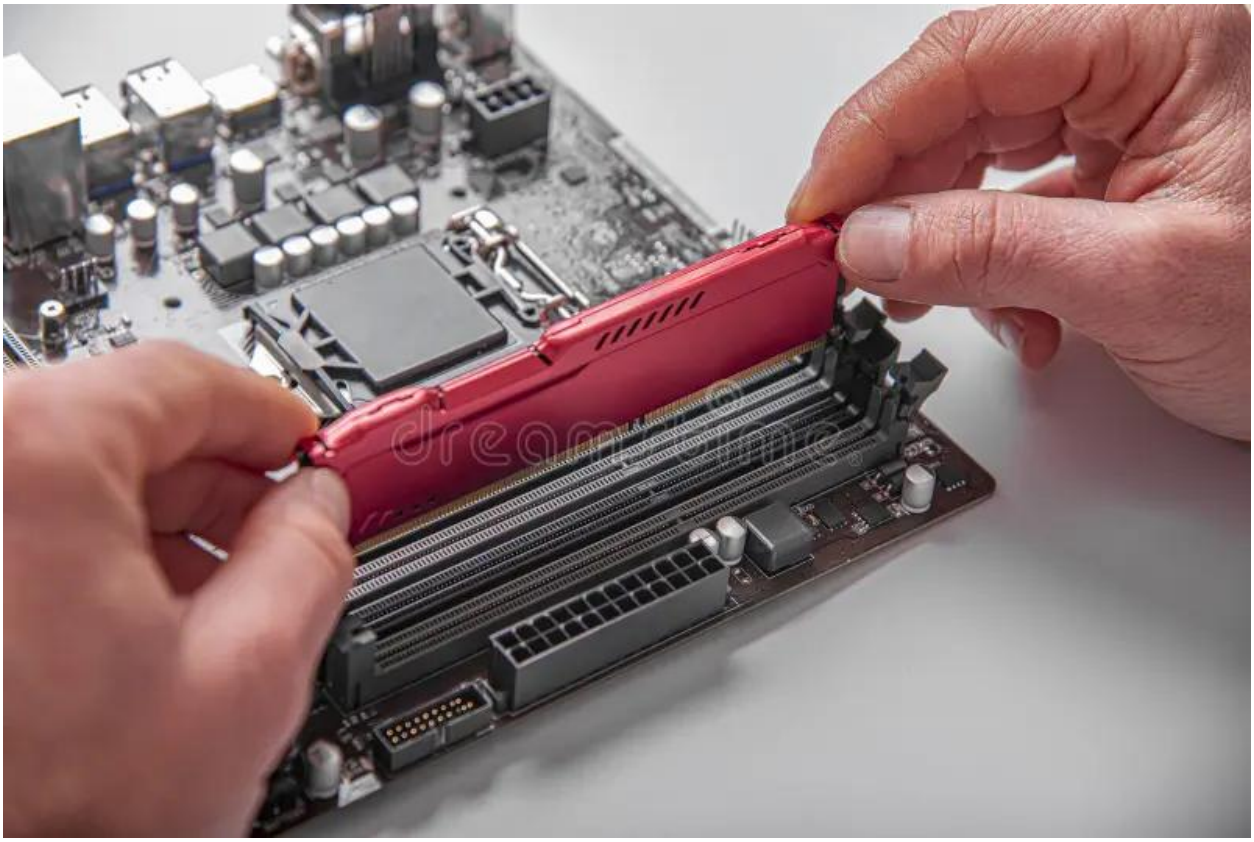
- Motherboard
- CPU and CPU Cooler
- RAM
- HDD/SSD
- SMPS
- SATA Cables
- Cooling Fans

Step 4: Disconnect Power and Data Cables

Power cables from the SMPS and SATA data cables connected to storage devices were carefully disconnected.

Step 5: Remove RAM Modules

The retaining clips on both sides of the RAM slot were pushed outward. The RAM module was lifted and removed carefully.



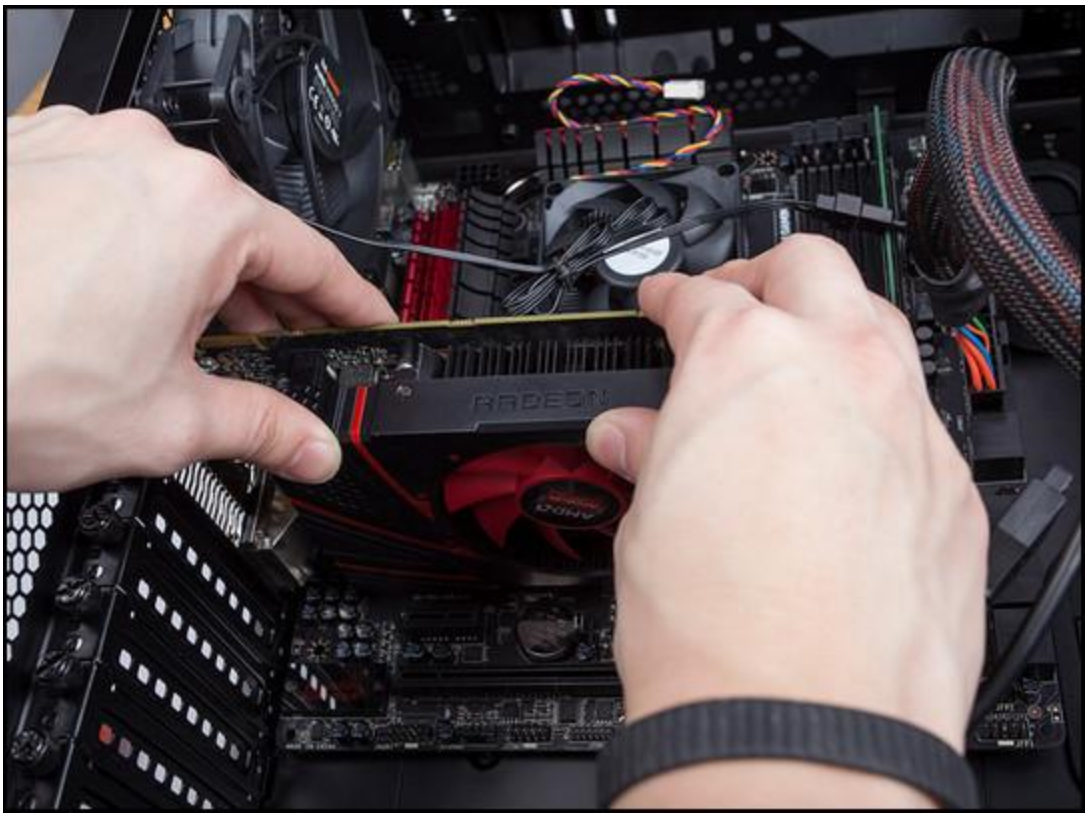
Step 6: Remove HDD/SSD

Mounting screws were removed and the storage drive was disconnected from power and SATA cables. The HDD/SSD was then removed from the drive bay.



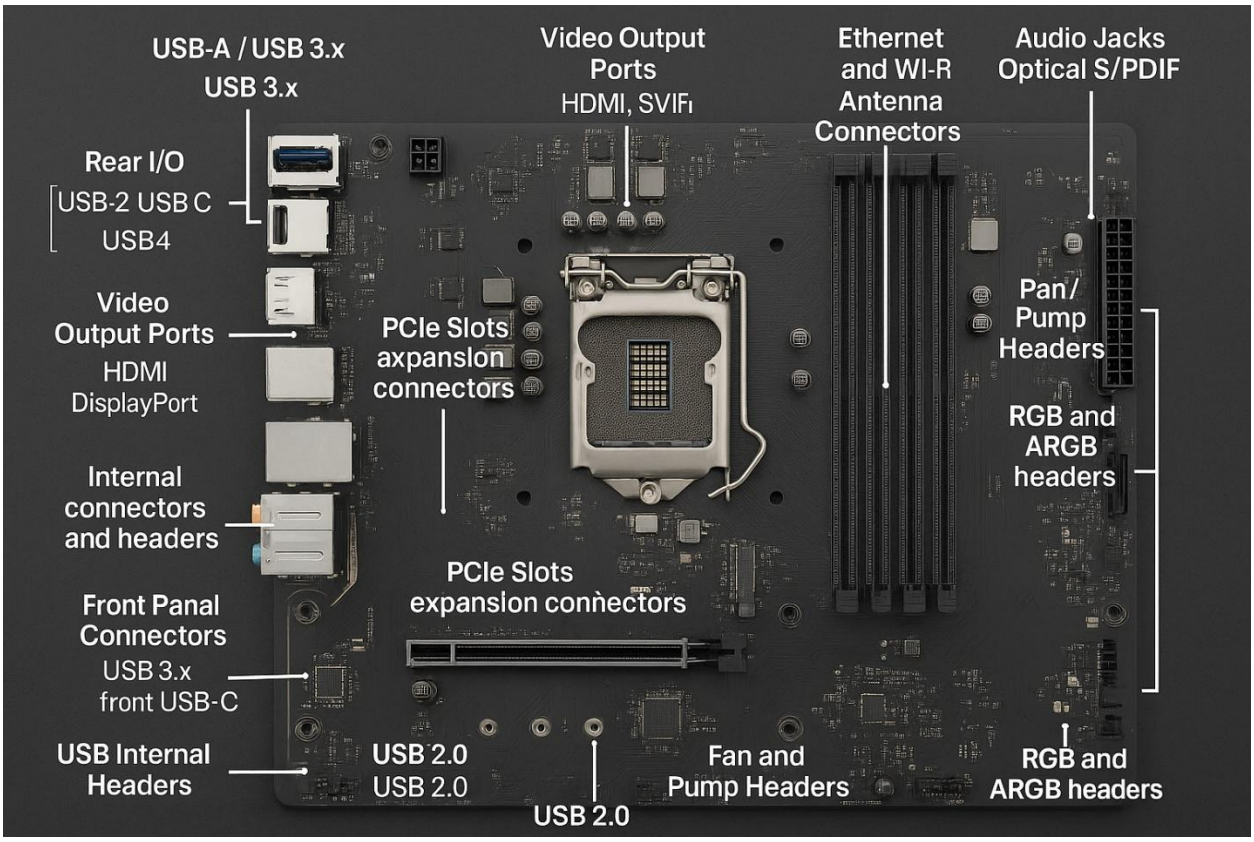
Step 7: Remove GPU

Action Performed:
Expansion card screws were removed and the cards were carefully pulled from their PCIe slots.



Step 8: Disconnect Motherboard Connections

Action Performed:
Front panel connectors, CPU power connector, motherboard power connector, USB headers, and fan cables were disconnected



Step 9: Remove the Motherboard

The mounting screws securing the motherboard to the computer case were removed. The motherboard was then carefully lifted from the cabinet.



Step 10: Final Inspection
All components were inspected for dust accumulation, physical damage, and loose connections.

3) Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

PC Reassembly Report

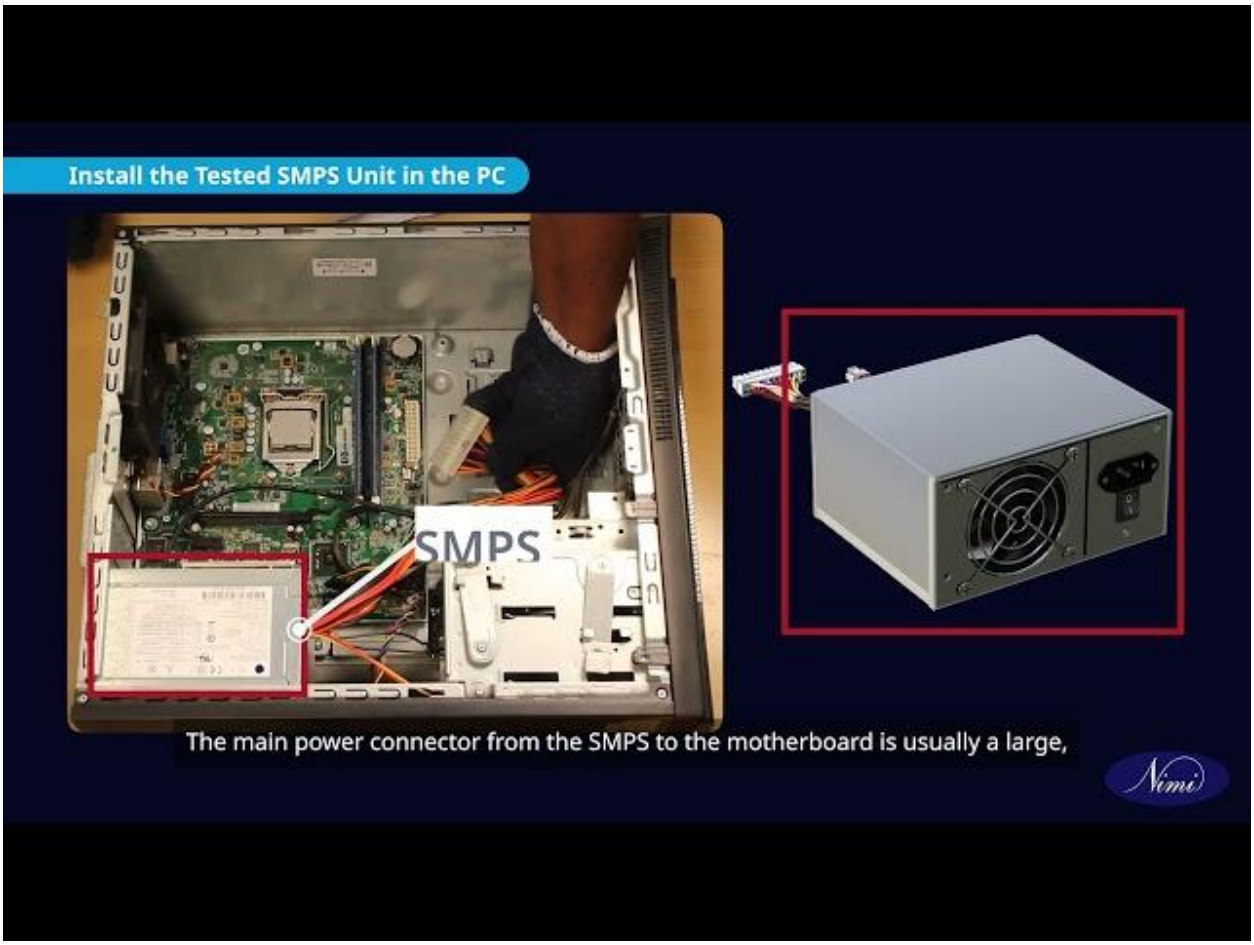
Step 1: Install the Motherboard

Action Performed:
The motherboard was carefully aligned with the standoffs inside the cabinet and secured using mounting screws.



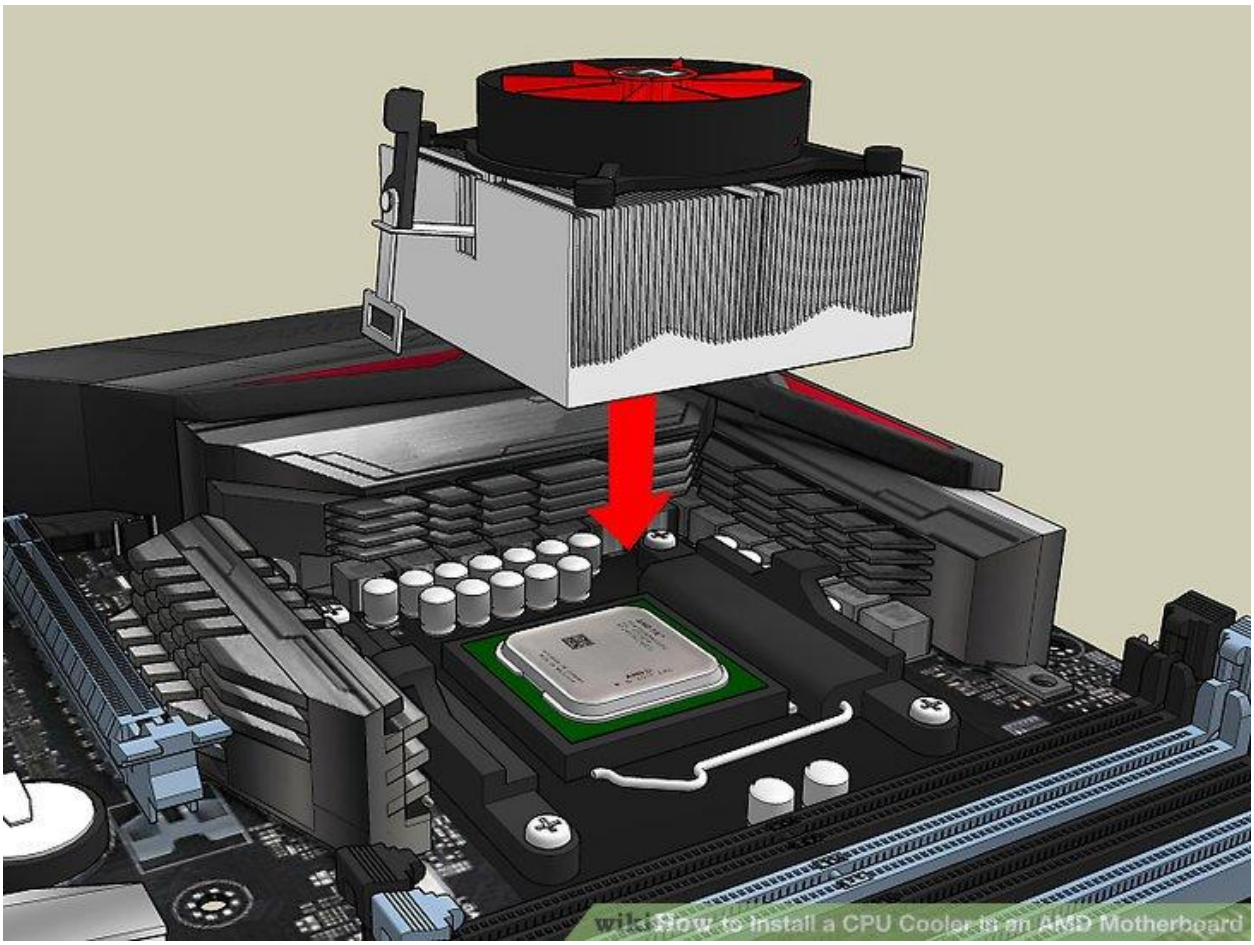
Step 2: Connect the SMPS (Power Supply)

Action Performed:
The SMPS was mounted in its designated compartment and secured with screws.



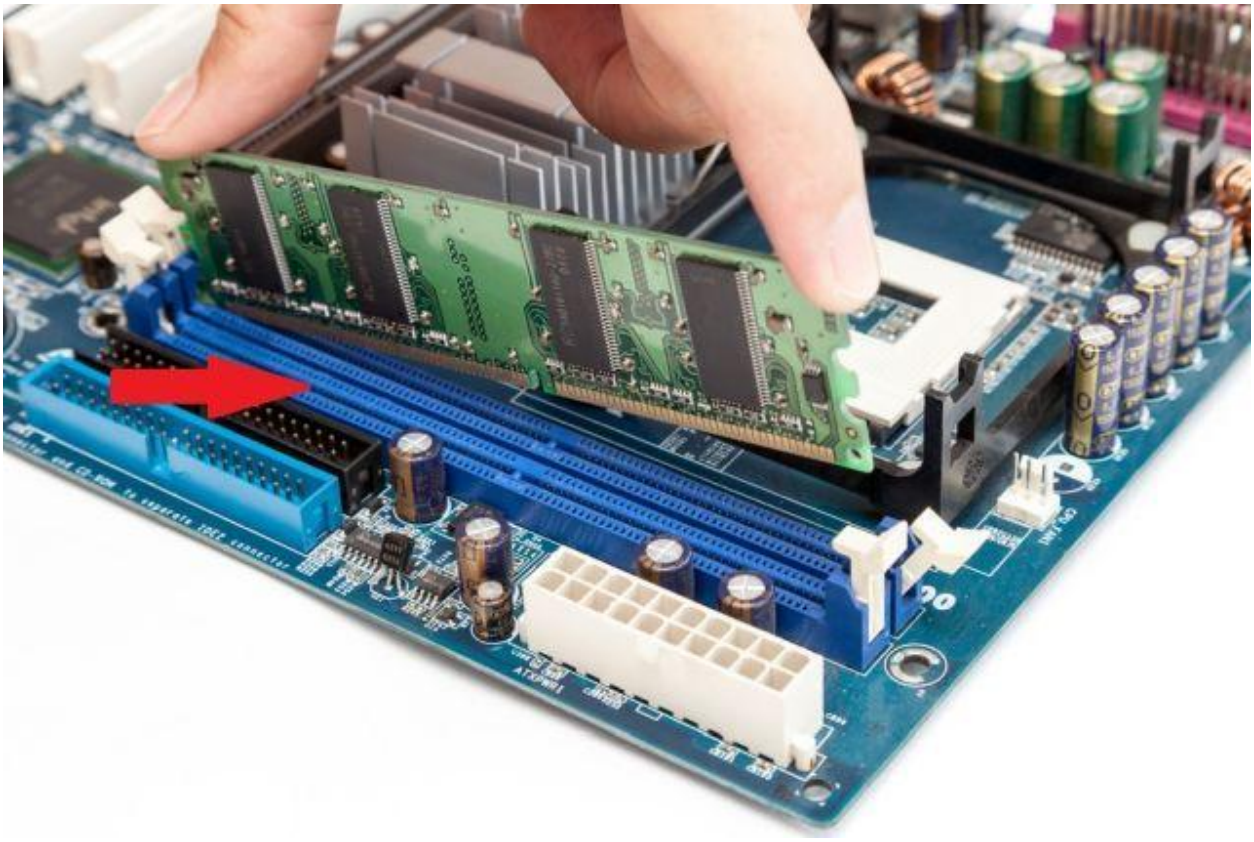
Step 3: Install the CPU and CPU Cooler

Action Performed:
The CPU was carefully placed into the motherboard socket and locked into position. The CPU cooler was then attached and connected to the CPU fan header.



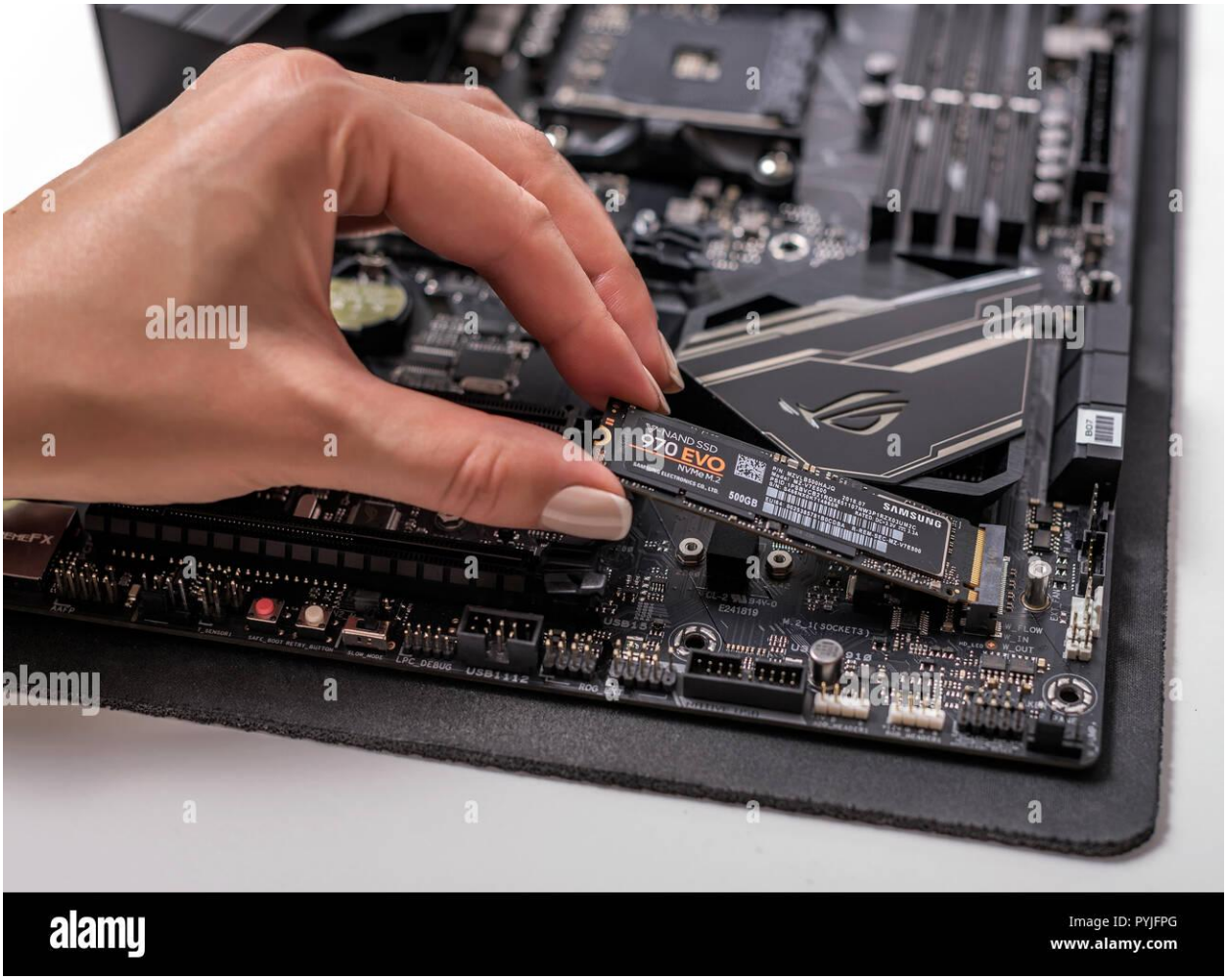
Step 4: Install the RAM Modules

Action Performed:
The RAM modules were aligned with the memory slots and pressed firmly until the retaining clips locked into place.



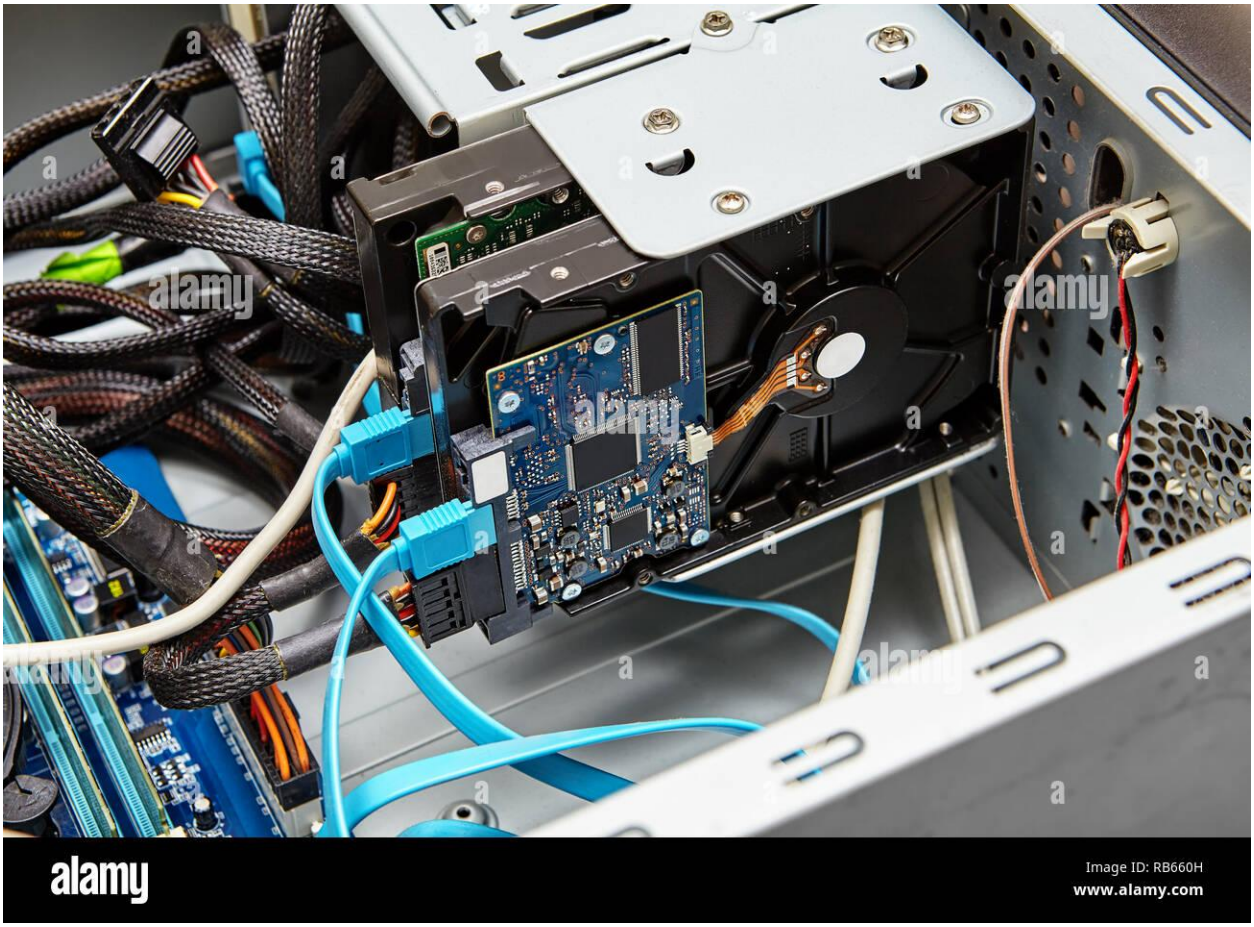
Step 5: Install HDD/SSD Storage Devices

Action Performed:
The HDD and SSD were placed in their drive bays and secured using screws



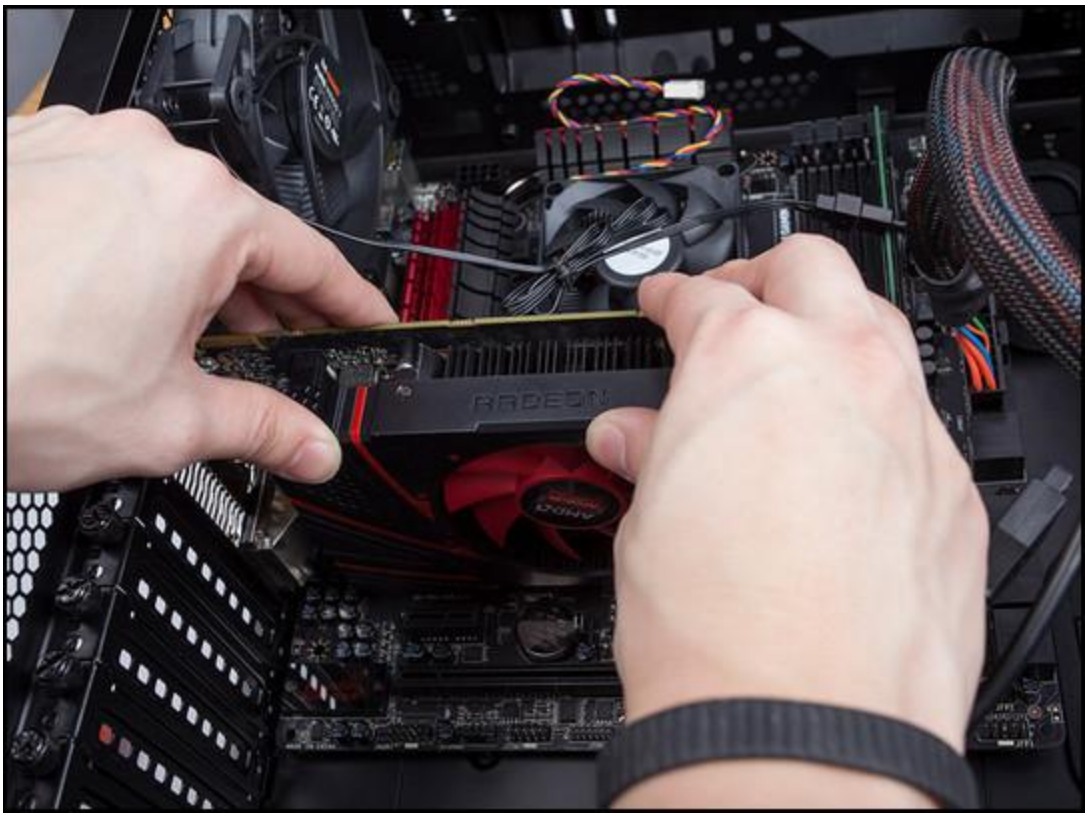
Step 6: Connect SATA Data and Power Cables

Action Performed:
SATA data cables were connected from the motherboard to the storage devices, and power cables from the SMPS were attached.



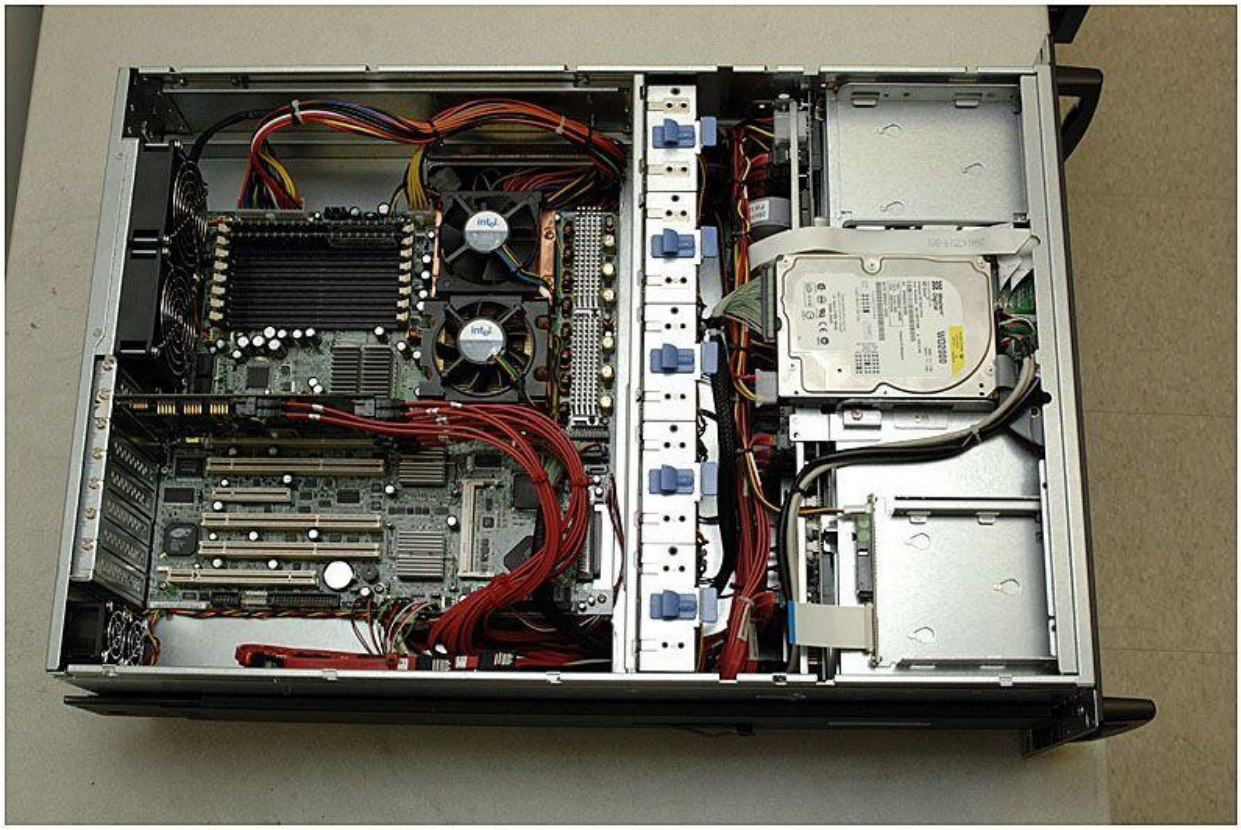
Step 7: Install the Graphics Card (GPU)

Action Performed:
The graphics card was inserted into the PCIe slot and secured with screws.



Step 8 : Attach the cables

Such as power cable,sata cable,usb cable



Step 9 : Power On and Verify Operation

Action Performed:
The power cable, monitor, keyboard, and mouse were reconnected. The system was powered on and checked for successful boot-up.

4) Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

Here is a comprehensive checklist of the essential tools and safety precautions required before starting any PC assembly or disassembly process, along with the reasons why each is crucial.

🛠️ Tools Checklist & Importance

- **Screwdriver (#2)**
    - **Why it's important:** Most screws used in PC cases, motherboards, and power supplies are standard Phillips screws. Using the correct size (#2) prevents stripping the screw heads or damaging surrounding components. A magnetized tip is especially useful for holding small screws in tight spaces.
- **Small Container or Magnetic Tray**
    - **Why it's important:** PC building involves dozens of tiny screws of varying lengths. A compartmentalized tray or magnetic bowl keeps them organized and prevents losing them inside the chassis or workspace.
- **Cable Ties / Velcro Straps**
    - **Why it's important:** Crucial during reassembly for routing cables neatly behind the motherboard tray, ensuring optimal airflow and preventing cables from sagging into spinning fans.
- **Isopropyl Alcohol (90% or higher) & Lint-Free Microfiber Cloth**
    - **Why it's important:** Essential for thoroughly cleaning old thermal paste off the CPU and cooler block before applying fresh paste during reassembly. High-purity alcohol evaporates quickly without leaving residue.

⚠️ Safety Precautions Checklist & Importance

- **Unplug the Power Cord and Switch Off the PSU**
    - **Why it's important:** Working on active hardware carrying live AC mains voltage or DC power risks severe electrical shock, component short circuits, and permanent hardware failure. Always switch off the Power Supply Unit (PSU) rocker switch and pull the plug from the wall outlet first.
- **Drain Residual Electrical Charge**
    - **Why it's important:** Even after unplugging the system, capacitors on the motherboard and power supply retain stored electrical energy. Pressing and holding the PC case's power button for 5 to 10 seconds discharges this residual electricity safely.
- **Ground Yourself Properly**

    - **Why it's important:** Electrostatic discharge (ESD) can occur invisibly with as little as 30 volts, whereas humans can feel shocks starting around 3,000 volts. Wearing an anti-static wrist strap clipped to a grounded metal object protects delicate microchips.

- **Work on a Stable, Non-Conductive Surface**
    - **Why it's important:** Avoid working on carpets, blankets, or conductive metal sheets that generate static or create accidental electrical shorts. A wooden workbench or rubberized mat is ideal.
- **Handle Components by Their Edges**
    - **Why it's important:** Touching gold contact pins (on RAM, GPUs, or CPUs) or exposed solder points on circuit boards transfers natural skin oils, dirt, and sweat, which can corrode contacts over time or cause short circuits. Always hold components by their edges or plastic shrouds.